Creator Defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach explanation and evidence as the defence criteria.

Supply complete rehearsal records, identify packet gaps and assess fresh figures before feedback.

Keep one habit: Ask which record supports the claim, which outcome is measured and which question the evidence cannot answer.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Explain the **mechanism** and terminology of generative systems **defense**

Two 45-minute sessions. A generative-system **defense** presents the question, **architecture**, corpus, permissions, baseline, evaluation and unresolved failures. Demonstration fluency is only one observable property.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and **limitation**s; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can explain creator **defense** using a traced example and a stated **limitation**.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Lesson and workbook; paper and pencil
- Calculator and optional local Python 3 for arithmetic; no live API or private data
- Prepare the fresh retry separately and withhold its solution until after collection

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Capstone Evaluate and Improve. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone evaluation and revision

Reference example, not the only acceptable wording: Teacher explanation: Capstone evaluation compares the frozen assistant with a baseline on reserved cases and preserves failures.

Revision uses observed errors but needs fresh **evidence** for new final claims. Report paired case outcomes, condition rates and permission violations separately.

A critical access failure can block release even when overall pass rate rises. Never overwrite the original final run.

Without opening the old lesson, explain one key idea from Capstone Design. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone requirements and corpus

Reference example, not the only acceptable wording: Teacher explanation: A grounded-assistant capstone begins with bounded requirements and a versioned permitted corpus.

A baseline can be a manual lookup **process** using the same **evidence**. Specify the user question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation.

Keep teacher material out of student retrieval.

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of representation and bias evaluation

Reference example, not the only acceptable wording: Teacher explanation: Bias evaluation tests defined outcomes across relevant conditions.

Group averages can expose differences but cannot alone establish their cause or the correct remedy. Define the task, relevant groups, labels and denominators before comparing. Use fictional, non-identifying records.

Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Capstone Evaluate and Improve

Capstone Design

Bias and Representation

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Your assistant passes 9/12 final questions, including all six direct facts but only 3/6 ambiguous or absent-**evidence** cases. Report results and argue for a focused next step.

Concept to explain

A creator explains **architecture**, **evidence**, **limitation**s, provenance, and responsible use.

Term	Meaning
defense	An explanation of a project decision supported by evidence and an account of its limitations.
architecture	The arrangement of a system's parts and the connections between them.
limitation	A condition or missing capability that restricts what a result can establish or a system can safely do.
transfer test	A fresh task that checks whether learning can be applied without the original supports.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach defensible claims with the matching units and records

> Accept an explicit evidence gap when the packet does not supply the answer.

State the defence evidence contract

Defence claim	Matching evidence and limitation
User value	Defined task and baseline comparison; no invented benefit
Architecture	Source/corpus versions, stages, permissions and owner map
Retrieval	Defined relevant-retrieval outcome and its own denominator
Answer support	Claim-level rubric and supported-answer denominator
Fallback	Missing-source cases and appropriate behavior rate
Actions and privacy	Actual recorded scope/trace; a zero in one run is not universal protection
Speed/resources	Timing/counts/tariff only when measured and defined
Reproduction	Exact versions/cases/outputs where available; mark missing packet fields
Fresh challenge	Predict, explain and record support; do not rewrite earlier evidence

A generative-system **defense** presents the question, **architecture**, corpus, permissions, baseline, evaluation and unresolved failures. Demonstration fluency is only one observable property.

Link each claim to a recorded result. State the distinction between retrieval relevance, claim support, fallback and permissions. Give a **reproduction packet** and answer challenges without inventing missing measurements.

Trace separate W2 relevance/support/action figures

System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run. Presenter claims 90% fully correct and faster than baseline.

Retrieval rate 90% and support rate **70%** differ. Action trace supports zero actions in that run; no timing **evidence** supports speed. Correct the full-correctness and speed claims.

Read complete D1 scope

Defend a study-buddy project using separate claims and matching **evidence**.

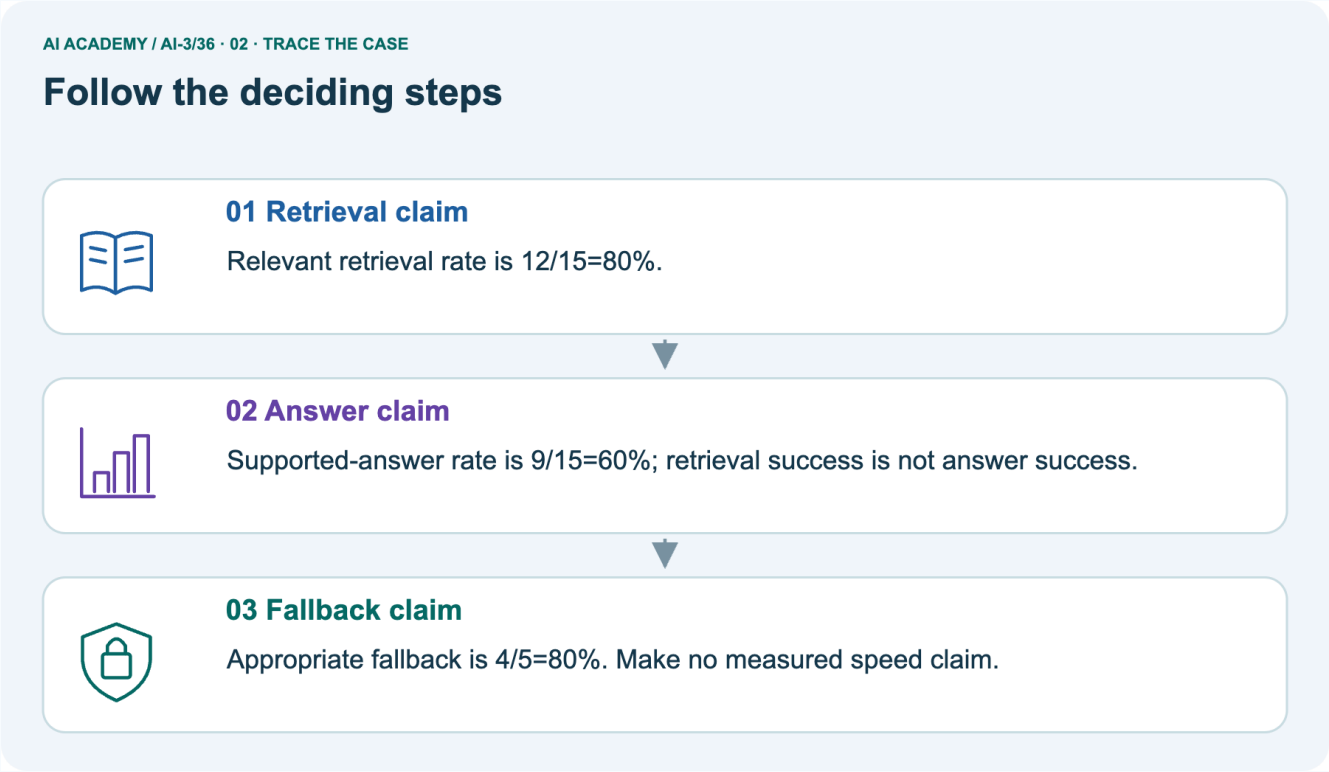
Set A: 15 answerable questions, with 12 relevant retrievals and 9 supported answers. Separate set B: 5 missing-source questions, with 4 appropriate fallbacks. No speed measurements exist.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Practice packet D1 supplies aggregate counts only: 15 answerable questions, 12 with relevant retrieval and 9 with supported answers; separate 5 missing-source questions with 4 appropriate fallbacks. Treat retrieval as a per-question pass indicator in this fixture, not document precision@k. Rates 80%, 60%, 80% answer different questions.

No timing, energy, case-level failure outputs or complete reproduction packet is supplied. Do not invent a specific failure explanation or a speed comparison from these totals.

Trace the three rates



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieval claim	Relevant retrieval rate is 12/15=80%.
2. Answer claim	Supported-answer rate is 9/15=60%; retrieval success is not answer success.
3. Fallback claim	Appropriate fallback is 4/5=80%. Make no measured speed claim.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Appropriate fallback is 4/5=**80%**. Make no measured speed claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the retrieval-only poster

Change: Only the retrieval rate is put on the presentation poster.

Prediction: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers.

Reason: Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

State fictional and incomplete-packet limits

These invented counts illustrate **defense** structure; the class must replace them with its own documented **evidence**.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Additional discussion · Never-hallucinates overclaim

Ten successful demos lead to a never hallucinates claim.

Write a defensible conclusion.

Report those ten specified passes, show **evidence** and fallback, and state untested conditions. Demonstrate an unanswerable question.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Trace one question through retrieval, generation, validation, and citation.

2. Recompute one evaluation score and show source support.

3. Run an unseen missing-source or conflicting-source case.

4. State intended use, do-not-use case, privacy/provenance controls, and human responsibility.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Exact explanation to model

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Defense claim	A statement tied to inspectable project evidence
Reproduction packet	Artifacts needed to repeat a reported procedure
Unmeasured outcome	A property for which the experiment supplies no observation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

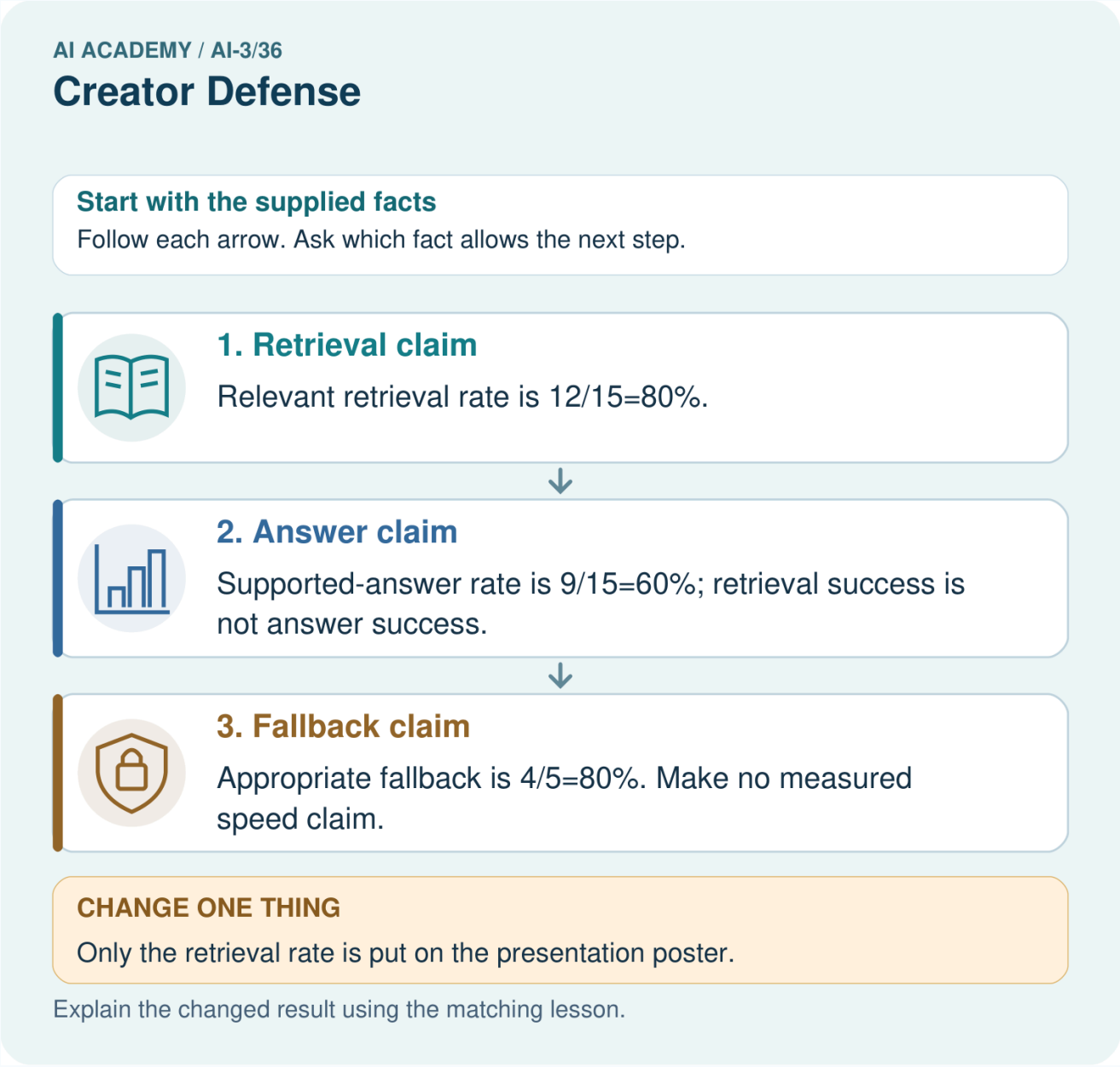
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Creator Defense

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the explanation, then remove help

> Collect each learner's reasoning before showing feedback.

Main completion prompt

Completion task: System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run. Presenter claims 90% fully correct and faster than baseline.

Show the intermediate steps and give the conclusion with its scope.

Expected reasoning and fading help

Teacher check: Teacher explanation: Retrieval rate 90% and support rate **70%** differ. Action trace supports zero actions in that run; no timing **evidence** supports speed. Correct the full-correctness and speed claims.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Generative systems **defense**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A **defense** should preserve negative **evidence** and uncertainty rather than treating polished presentation as proof.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Which numbers would you display to make the **defense** honest? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Display 12/15 retrieval, 9/15 support and 4/5 fallback with separate set A and set B definitions; say speed was not measured.

Changed-case check: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers. Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

Changed-case reasoning: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers. Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ **Earlier wording, alternatives and picture transcripts**

Demonstrate and guide practice

Board sequence

Write the given facts: Ten successful demos lead to a never hallucinates claim.

Write the question to explain: Write a defensible conclusion.

Reveal after **prediction**s: Report those ten specified passes, show **evidence** and fallback, and state untested conditions. Demonstrate an unanswerable question.

Use the earlier changed question as optional later practice. Reserve Defend the fresh outcome figures independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: Retrieval rate 90% and support rate **70%** differ. Action trace supports zero actions in that run; no timing **evidence** supports speed. Correct the full-correctness and speed claims.

Qualification: A **defense** should preserve negative **evidence** and uncertainty rather than treating polished presentation as proof.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage	Teacher action
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

D1 rates 80% retrieval/60% support/80% fallback with different case roles; no speed or complete reproduction packet. D2 day Tuesday via C1-v1, leader absent. Do not invent six failure types from aggregate counts. W2 retrieval 90%, support 70%, zero actions only in supplied run. Independent 80%/60%/75% with denominators 10/10/4; no latency or universal correctness claim.

Defend a bounded record rather than a perfect story.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the complete D1/D2 rehearsal packet and challenge cards. Present separate rates, trace a supplied supported/missing-source scenario, identify missing reproduction details and reject unmeasured speed claims. Label predictions, paper traces and actual project results separately; do not adopt fictional counts as student **evidence**.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

The worked result establishes every future case: A **defense** should preserve negative **evidence** and uncertainty rather than treating polished presentation as proof.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Defend the fresh outcome figures independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A **defense** has retrieval 8/10, support 6/10, fallback 3/4 and no measured latency. State the three rates and reject one unsupported claim.

Teacher explanation: **80%** retrieval, **60%** support, 75% appropriate fallback; denominators differ. Speed and universal correctness are unmeasured; present **limitation**s and source versions.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A generative-system **defense** presents the question, **architecture**, corpus, permissions, baseline, evaluation and unresolved failures. Demonstration fluency is only one observable property.

Link each claim to a recorded result. State the distinction between retrieval relevance, claim support, fallback and permissions. Give a **reproduction packet** and answer challenges without inventing missing measurements.

W2 Trace the worked case

System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run. Presenter claims 90% fully correct and faster than baseline.

Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Retrieval rate 90% and support rate **70%** differ. Action trace supports zero actions in that run; no timing **evidence** supports speed. Correct the full-correctness and speed claims.

W3 Audit the limit

A report omits this qualification: A **defense** should preserve negative **evidence** and uncertainty rather than treating polished presentation as proof.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A **defense** should preserve negative **evidence** and uncertainty rather than treating polished presentation as proof.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A **defense** should preserve negative **evidence** and uncertainty rather than treating polished presentation as proof.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Report those ten specified passes, show **evidence** and fallback, and state untested conditions. Demonstrate an unanswerable question.

Additional transfer practice question

An audience asks whether your assistant is always correct. Give an **evidence**-based **defense** using results and a limit.

Reasoned practice answer

Describe the evaluation tasks, results, failures and revisions; explain the source boundary and remaining uncertainty. Do not claim universal correctness from a limited test.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A successful capstone demonstration proves universal reliability.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Overall is 75%; direct facts 100%; difficult cases 50%. Prioritise clarification and abstention, retaining each failure record. Twelve questions do not establish broad reliability.

Demonstrate a correct unsupported-question response during the **defense**.

Explanations for the additional workbook examples

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

A **defense** has retrieval 8/10, support 6/10, fallback 3/4 and no measured latency. State the three rates and reject one unsupported claim.

Teacher explanation: **80%** retrieval, **60%** support, 75% appropriate fallback; denominators differ. Speed and universal correctness are unmeasured; present **limitation**s and source versions.

W5 Design a revealing follow-up

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Final product card documents purpose, system map, sources, evals, limits, risks, and human oversight.

Keep the decision, **input**, result, explanation and remaining **limitation** in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Verified Study Buddy, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: **80%** retrieval, **60%** support, 75% appropriate fallback; denominators differ. Speed and universal correctness are unmeasured; present **limitation**s and source versions. Require intermediate reasoning and the stated scope.

Fresh retry: Retrieval 6/8, support 5/8, fallback 2/3; no energy measurement. Expected reasoning: 75%, 62.5%, 66.667% with stated denominators; energy efficiency cannot be claimed.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Plan the delayed check

Suggested return points in this level: plan a review after the level ends. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Review the full project **evidence** and choose concepts needing another independent attempt before the next level. One expo is not proof of mastery of all AI.

After Level 3, use the Python bridge programme before Level 4. Carry forward explicit input contracts, source/data roles, reproducible code and honest test **evidence**. Revisit the capstone on a fresh case after the level ends; record support used.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Explain the **mechanism** and terminology of generative systems **defense**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Practice and assessment reminders

Before class, solve the supported practice: System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run. Presenter claims 90% fully correct and faster than baseline.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Retrieval rate 90% and support rate **70%** differ. Action trace supports zero actions in that run; no timing **evidence** supports speed. Correct the full-correctness and speed claims.

Check every learner before discussion: Use the worked case for Generative systems **defense**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A **defense** should preserve negative **evidence** and uncertainty rather than treating polished presentation as proof.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Only the retrieval rate is put on the presentation poster. Require a **prediction** and a reason before comparing.

Expected result: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers. Reason: Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and **limitation**s. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
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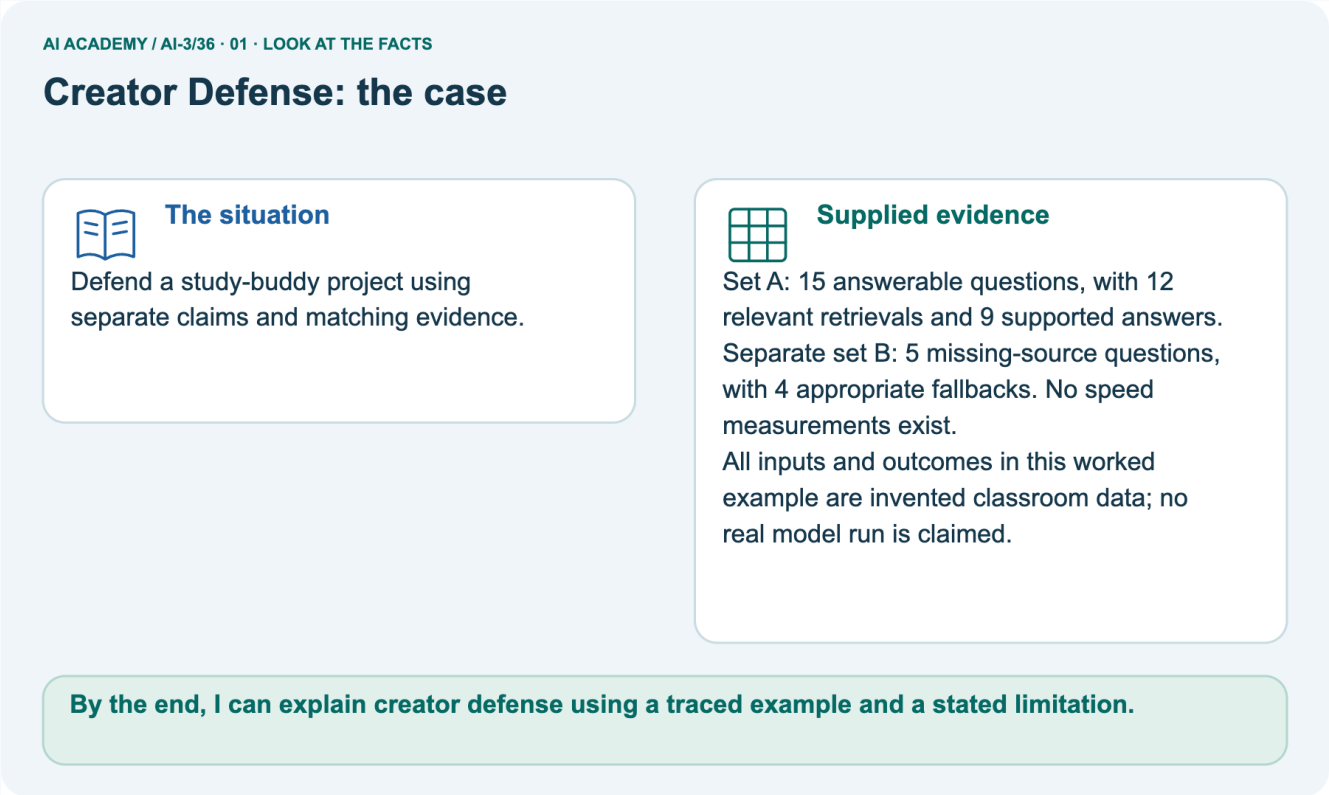
<details>
<summary>Reference: teach with the six pictures</summary>
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REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Creator Defense: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

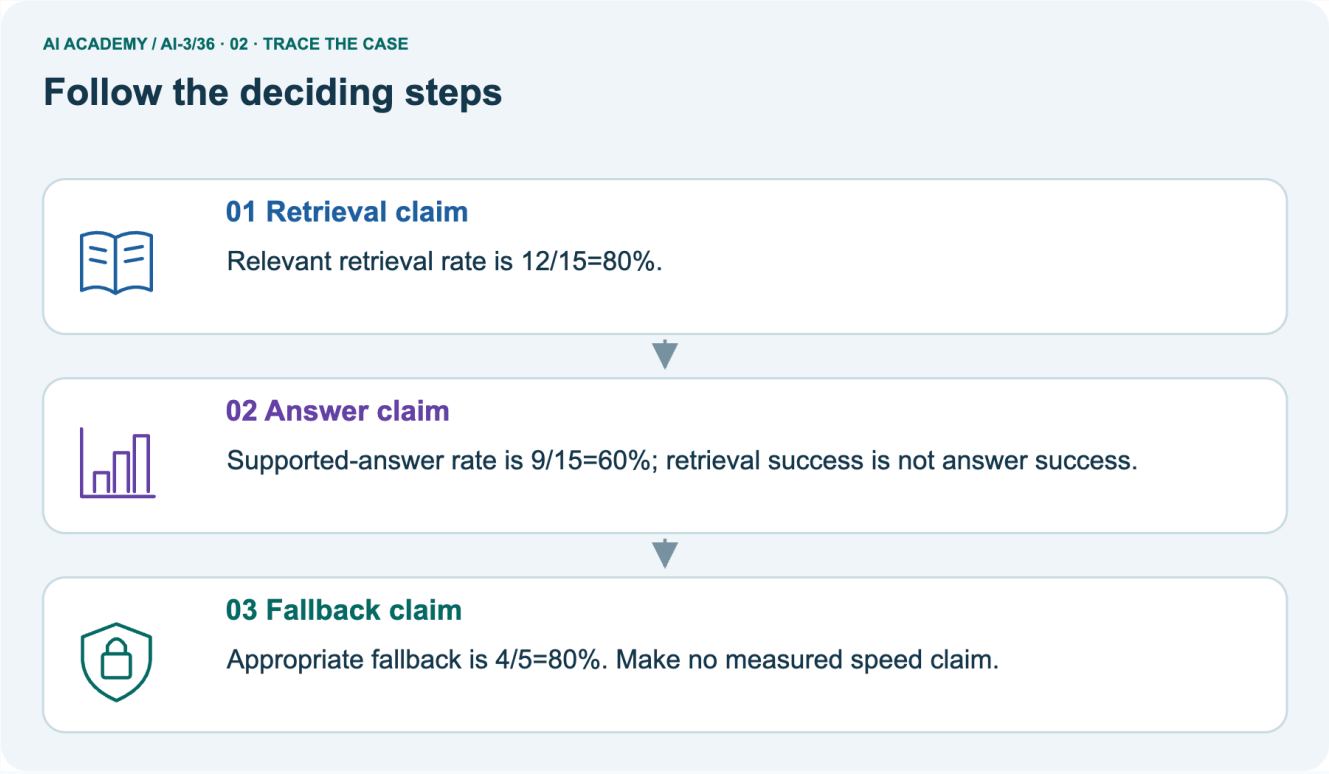
Defend a study-buddy project using separate claims and matching **evidence**. Supplied facts: Set A: 15 answerable questions, with 12 relevant retrievals and 9 supported answers.

Separate set B: 5 missing-source questions, with 4 appropriate fallbacks. No speed measurements exist. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

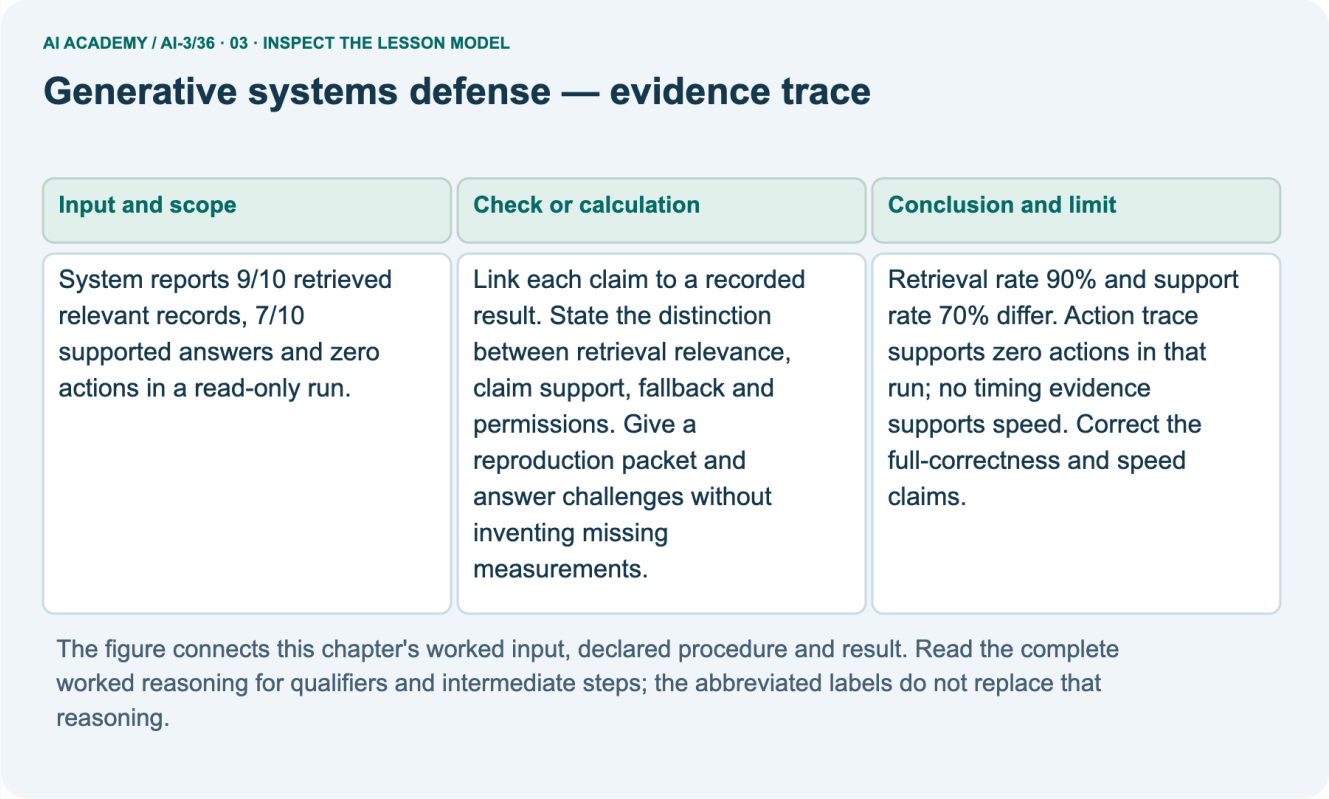
Retrieval claim: Relevant retrieval rate is 12/15=**80%**. → Answer claim: Supported-answer rate is 9/15=**60%**; retrieval success is not answer success. → Fallback claim: Appropriate fallback is 4/5=**80%**.

Make no measured speed claim.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Generative systems defense — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Generative systems **defense** — **evidence** trace. System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run. | Link each claim to a recorded result.

State the distinction between retrieval relevance, claim support, fallback and permissions.

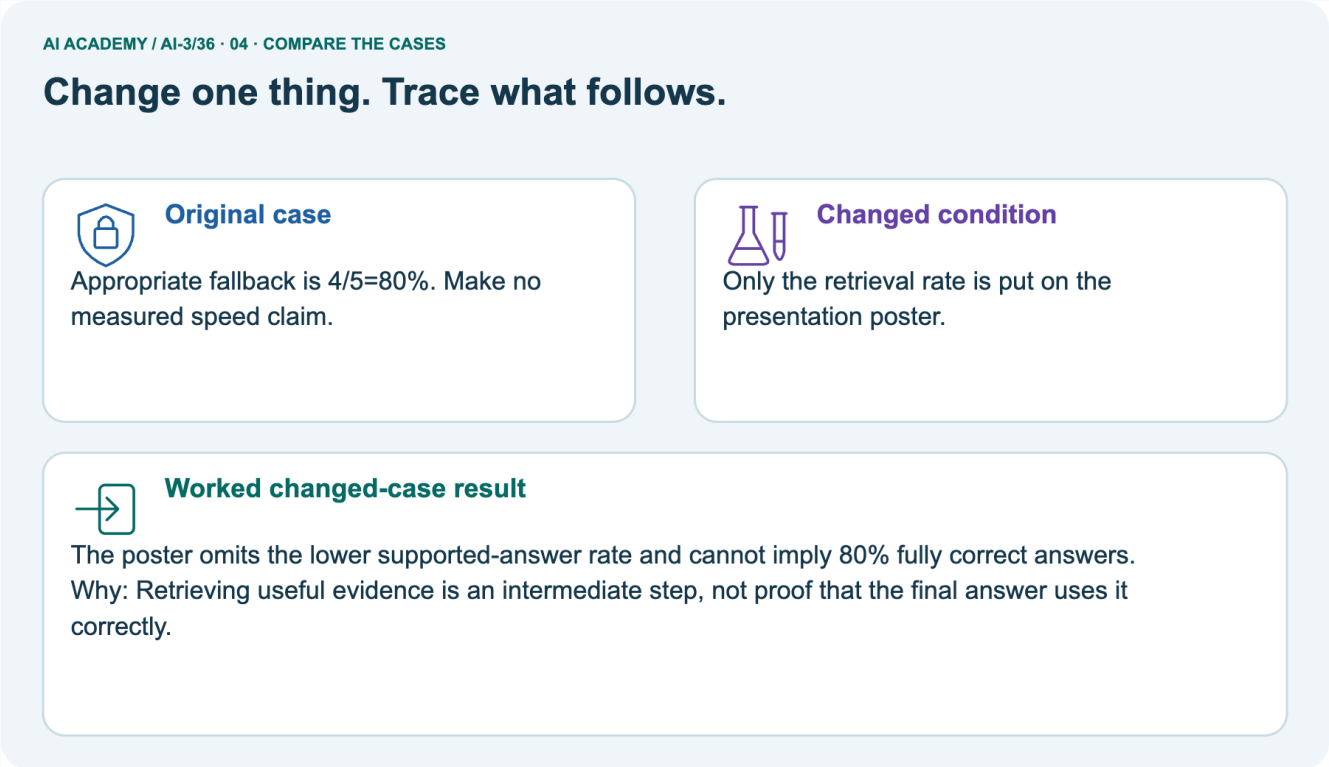
Give a **reproduction packet** and answer challenges without inventing missing measurements. | Retrieval rate 90% and support rate **70%** differ. Action trace supports zero actions in that run; no timing **evidence** supports speed.

Correct the full-correctness and speed claims.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Appropriate fallback is 4/5=**80%**. Make no measured speed claim. Changed condition: Only the retrieval rate is put on the presentation poster.

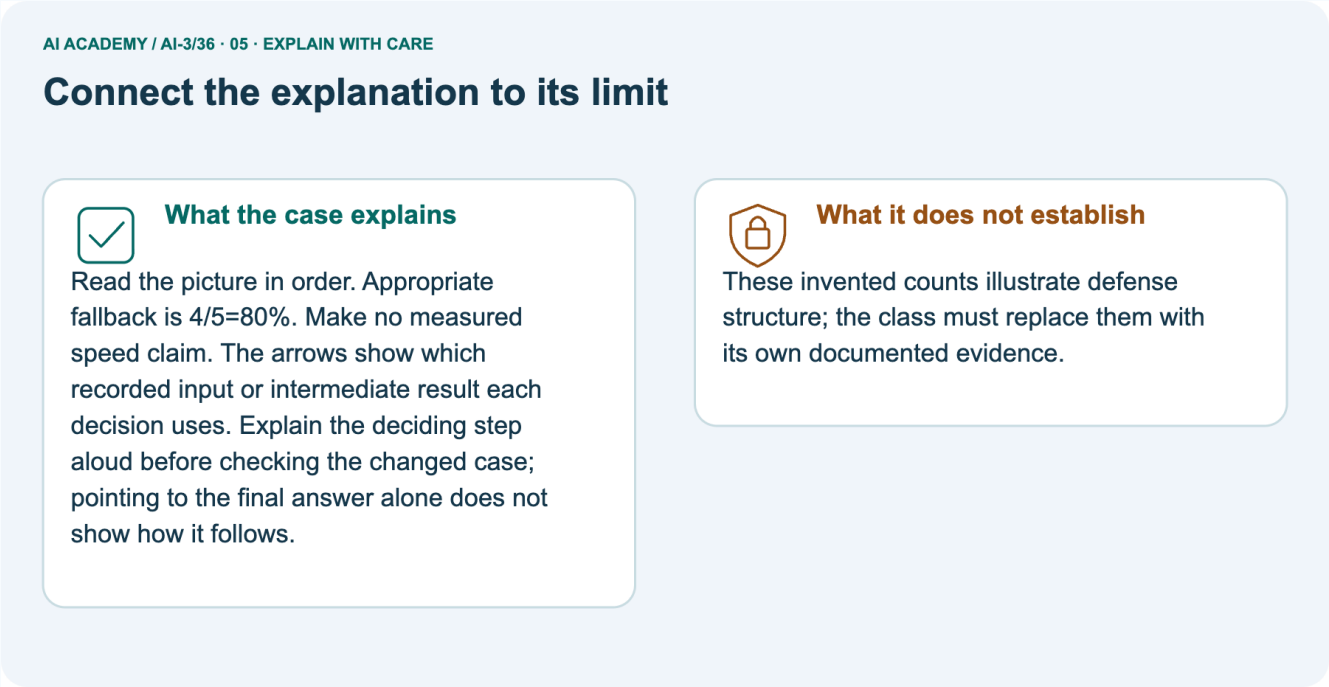
Expected result: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers. Reason: Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Appropriate fallback is 4/5=**80%**. Make no measured speed claim. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

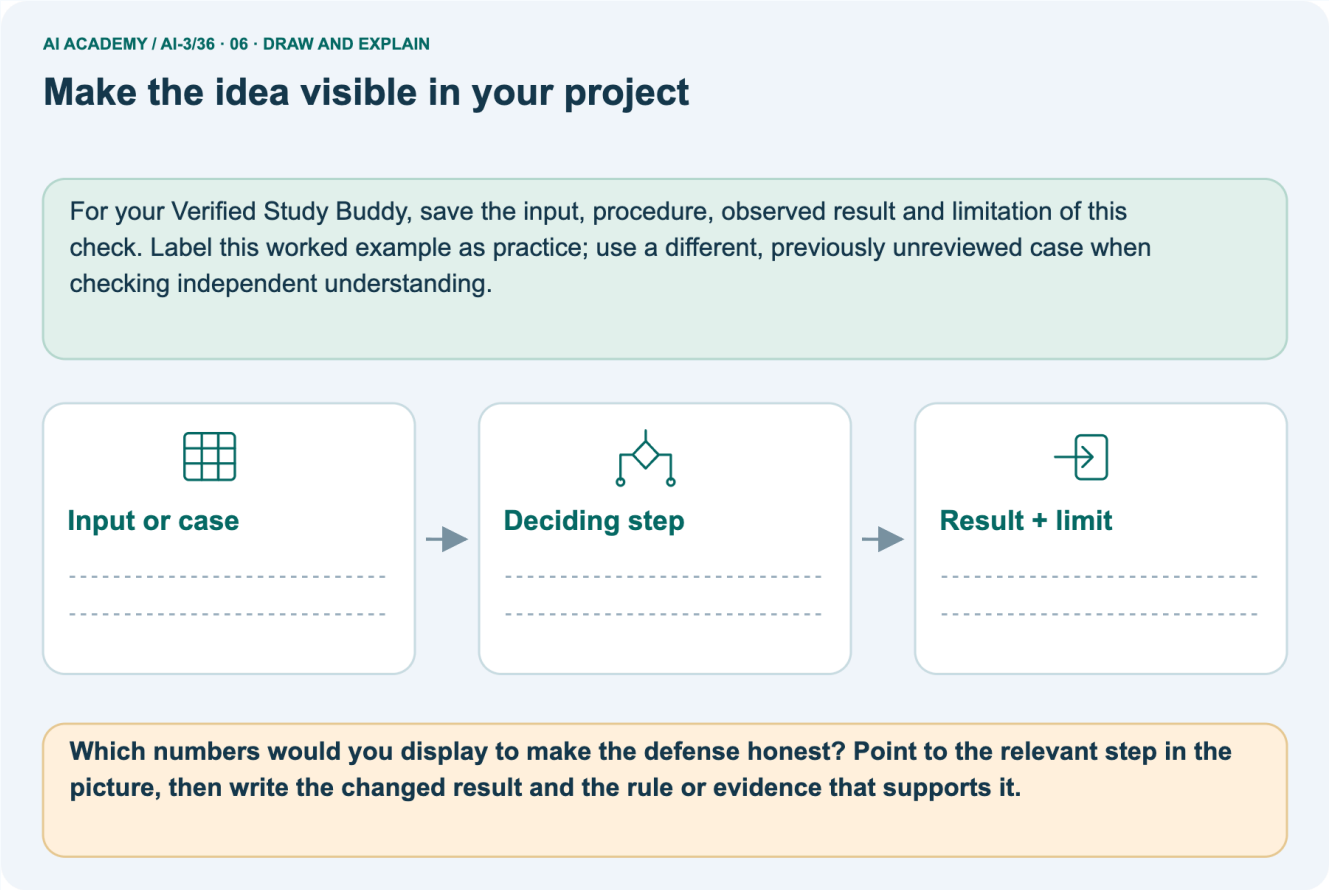
Limit: These invented counts illustrate **defense** structure; the class must replace them with its own documented **evidence**.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These invented counts illustrate **defense** structure; the class must replace them with its own documented **evidence**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which numbers would you display to make the **defense** honest? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Display 12/15 retrieval, 9/15 support and 4/5 fallback with separate set A and set B definitions; say speed was not measured.

Changed-case check: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers. Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Creator Defense: the case

02. Follow the deciding steps

03. Generative systems defense — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.